Department für Wirtschafts- und Rechtswissenschaften

Masterarbeit

Demand Estimation with Unstructured Product Data: Evidence from Amazon's Toothpaste Market

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Danksagung

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Erklärung

Ich erkläre, dass ich diese Arbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel verwendet habe.

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Abstract

Merger simulation hinges on correctly measuring substitution patterns, yet the nesting structures used in practice are typically built from coarse labels such as brand identity. This thesis asks whether product representations extracted from unstructured listing content – images and text encoded with CLIP – define substitution patterns better than brand labels do. Using an ASIN-by-quarter panel of the Amazon US toothpaste market around Colgate-Palmolive's 2020 acquisition of the natural-brand hello, three nested-logit demand models are estimated via the Berry (1994) linearisation that differ only in how nests are assigned: no nesting, brand nesting, and nesting by k-means clusters of CLIP image-and-text embeddings. CLIP-based nesting fits demand best and cuts the share of economically implausible negative implied marginal costs from 8.6% to 1.7%. Because CLIP places hello and Colgate in different clusters, the implied diversion between the merging parties is small: the simulated merger price effect for hello falls from +1.23% under plain logit to +0.40% under CLIP nesting. Placebo tests confirm that any cross-firm nesting can repair the cost side; what is distinctive to CLIP is that it also fits demand better than 97.5% of random partitions and yields a merger counterfactual free of the spurious diversion those partitions introduce. The results suggest embedding-based nests are a practical, replicable tool for antitrust analysis when conventional product characteristics are unavailable.

Zusammenfassung

Fusionssimulationen hängen entscheidend davon ab, Substitutionsmuster korrekt zu messen; die in der Praxis verwendeten Nest-Strukturen beruhen jedoch meist auf groben Merkmalen wie der Markenzugehörigkeit. Diese Arbeit untersucht, ob Produktrepräsentationen aus unstrukturierten Angebotsinhalten – mit CLIP kodierte Bilder und Texte – Substitutionsmuster besser abbilden als Markenlabels. Auf Basis eines ASIN-Quartals-Panels des US-amerikanischen Amazon-Zahnpastamarktes rund um die Übernahme der Naturkosmetikmarke hello durch Colgate-Palmolive im Jahr 2020 werden drei Nested-Logit-Nachfragemodelle mittels der Berry-(1994)-Linearisierung geschätzt, die sich ausschließlich in der Nest-Zuordnung unterscheiden: ohne Nester, Marken-Nester und Nester aus k-Means-Clustern der CLIP-Einbettungen. Die CLIP-basierte Nest-Struktur weist die beste Anpassung auf und senkt den Anteil ökonomisch unplausibler negativer implizierter Grenzkosten von 8,6 % auf 1,7 %. Da CLIP hello und Colgate unterschiedlichen Clustern zuordnet, ist die implizierte Diversion zwischen den Fusionsparteien gering: Der simulierte Preiseffekt der Fusion für hello sinkt von +1,23 % (einfaches Logit) auf +0,40 % (CLIP-Nester). Placebo-Tests bestätigen, dass jede firmenübergreifende Nest-Struktur die Kostenseite reparieren kann; CLIP zeichnet sich dadurch aus, dass es zusätzlich die Nachfrage besser anpasst als 97,5 % zufälliger Partitionen und eine Fusionssimulation liefert, die frei von der durch solche Partitionen erzeugten Scheindiversion ist. Die Ergebnisse legen nahe, dass einbettungsbasierte Nester ein praktikables, reproduzierbares Instrument der Wettbewerbsanalyse sind, wenn klassische Produkteigenschaften nicht verfügbar sind.

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1 Introduction

Antitrust authorities increasingly confront mergers in digital markets, typically an established incumbent buying a fast-growing digital-native brand, where product differentiation is not captured by standardised attributes. In a car merger the analyst observes horsepower and fuel economy; in a laptop merger, RAM and screen size. Online consumer goods are different. There the differentiation that matters – a "natural" positioning, sensitivity relief, a kid-friendly format – lives in unstructured listing content: product images, titles, and free-form benefit descriptions. The workhorse tools of merger simulation were built for the structured world. Nested-logit models (McFadden 1978) encode substitution through nest assignments constructed from coarse observable labels, usually brand identity or hand-picked segments; when those labels misrepresent how consumers perceive products, the resulting diversion ratios, markups, and simulated price effects inherit the error.

A second, equally practical obstacle compounds the first. Estimating price sensitivity credibly requires exogenous cost-side variation, and high-frequency online retail data rarely contain any: input-cost series vary only over time and are absorbed by time fixed effects, while Hausman-style rival-price instruments (Hausman 1996) fail when demand shocks are correlated across markets, as they plainly are on a single national platform. The symptom of proceeding without valid instruments is well known to practitioners: demand estimated far too inelastic, implying Bertrand markups so large that the marginal costs recovered from the first-order conditions turn *negative* for a substantial share of products. The negative-cost rate directly measures how badly a demand system describes competition, and it is the diagnostic this paper puts at its centre.

This paper offers a pragmatic pipeline for exactly that environment, with one methodological contribution at its core. The pipeline maps unstructured product listings into a continuous differentiation space using a pre-trained deep learning model: CLIP image and text embeddings (Radford et al. 2021), reduced to five joint principal components. It sidesteps the missing-instrument problem by calibrating the price coefficient to an established industry elasticity, the meta-analytic mean own-price elasticity of -2.62 for consumer packaged goods (Bijmolt, van Heerde, and Pieters 2005). The contribution lies in what the embeddings are used *for*: nests come from k-means clustering in the embedding space, not from brand. Holding everything else fixed, we compare three Berry (1994) nested-logit models that differ only in their nest assignment (no nesting, brand nesting, and CLIP-cluster nesting) and let the negative-cost diagnostic decide between them.

The empirical laboratory is Colgate-Palmolive's acquisition of hello Products LLC, a fast-growing premium "naturally friendly" oral-care brand. The deal was announced on 23 January 2020 and completed on 31 January 2020 for \$351 million in cash, giving a clean event date (2020Q1) inside the sample. Using an ASIN-by-quarter panel of the Amazon US toothpaste market (2018Q1–2022Q3, 159 products), we estimate three demand models via the Berry (1994) nested-logit linearisation that are identical in every respect except the nest assignment: M1 imposes no nesting (plain logit), M2 nests by brand, and M3 nests by K=5 CLIP clusters. With the price coefficient fixed by calibration, the remaining parameters are estimated by concentrated OLS with quarter fixed effects and the nesting parameter ρ is profiled by grid search. Marginal costs are recovered from the Bertrand first-order conditions, and the merger is simulated by recomputing Nash equilibrium prices under pre- and post-merger ownership at fixed pre-merger costs.

Three results stand out. First, on fit: the profiled nesting parameter for CLIP nesting is $\rho^* = 0.67$, and the residual sum of squares falls by 8.1% relative to plain logit – brand nesting manages 3.1%. Second, and most important, the cost side. CLIP nesting drives the share of observations with *negative implied marginal costs* from 8.6% under both plain logit and brand nesting down to 1.7%. Negative recovered costs are the classic symptom of a demand system that understates substitutability and therefore overstates markups, so their near-elimination means the cost side of the model is repaired. Is that repair itself evidence the embeddings capture something real? Not by itself – a placebo analysis (Section 5.6) shows any cross-firm nesting achieves it. What is specific to CLIP is the pairing: demand fit better than 97.5% of random partitions, and a merger prediction no random partition reproduces. Which brings us to the third result. CLIP places hello and Colgate in different clusters, implying low diversion between the merging parties; the simulated price effect for hello falls from +1.23% (plain logit) to +0.40% (CLIP nesting), and Colgate's own effect is essentially zero.

Beyond the specific merger, the exercise is built as a transparent, reproducible template for antitrust practitioners. Every step, from embedding construction through the ρ grid search to the Nash counterfactuals, runs from a single document, and the Nash solver is validated against PyBLP's counterfactual routines (Conlon and Gortmaker 2020) to numerical precision. The remainder of the paper proceeds as follows. Section 2 relates the approach to the literature. Section 3 describes the industry setting, the data, and the embedding construction. Section 4 lays out the empirical framework. Section 5 presents the results, and Section 6 discusses limitations. Section 7 concludes.

2 Related Literature

Three literatures meet in this paper: the structural estimation of differentiated-products demand and its use in merger analysis, the calibration of price sensitivity when cost-side instruments are weak, and the newer use of machine-learned representations as economic product characteristics. The contribution is at their intersection.

2.1 Discrete-choice demand and merger simulation

The demand framework descends from the discrete-choice literature on differentiated products. Berry (1994) shows that logit and nested-logit demand can be estimated by linear methods after inverting market shares, and that the same first-order conditions used to rationalise observed prices can be inverted to recover marginal costs under Bertrand competition. Berry, Levinsohn, and Pakes (1995) generalise this framework to random-coefficients demand; Grigolon and Verboven (2014) show that a well-specified nested logit often approximates random-coefficients substitution patterns closely, which makes the choice of nests – the object studied here – the binding constraint in practice. In practice that choice rests on institutional priors: nests are drawn along brand, category, or a few observed characteristics, and the substitution pattern is only ever as good as those priors. Nevo (2000a) provides the standard practitioner's treatment of this class of models, and Nevo (2001) demonstrates the full structural program – demand estimates, recovered margins, and competitive counterfactuals – in the ready-to-eat cereal industry, a consumer packaged goods setting closely analogous to toothpaste.

Merger simulation itself follows Werden and Froeb (1994) and Nevo (2000b): recover marginal costs from the pre-merger equilibrium, change the ownership structure, and re-solve for Bertrand-Nash prices – computed here with the damped fixed-point iteration analysed by Morrow and Skerlos (2011). Conlon and Gortmaker (2020) codify current best practice and provide the PyBLP routines against which this paper's equilibrium computations are validated. Do simulations of this kind predict what actually happens after a merger? The retrospective evidence says: partly. Peters (2006) finds that standard simulations recover a large share of the observed price changes across six U.S. airline mergers, and Björnerstedt and Verboven (2016) find that nested-logit simulations correctly or moderately underpredict the 40% price increase the merging firms pushed through after a Swedish analgesics merger. Those studies check the method against realised prices. The placebo analysis of Section 5.6 brings the same spirit inside a single application – not predictions against realised prices, but the preferred nesting against the space of alternative structures.

2.2 Identification and the calibration of price sensitivity

Within this tradition, the price coefficient here is calibrated, not instrumented. The problem it steps around is a familiar one. Price moves with unobserved quality, so ordinary least squares on the share equation returns a coefficient biased toward zero – demand that looks too inelastic – and the usual fixes (cost shifters, BLP-style differentiation instruments, or Hausman's 1996 other-market prices) all lean on exclusion restrictions that are hard to defend in a thin, single-category panel on one national platform, for the reasons the introduction lays out. So rather than instrument badly, we fix the coefficient to an external target. That target, an elasticity of -2.62, comes from Bijmolt, van Heerde, and Pieters (2005), who meta-analyse 1,851 own-price elasticity estimates from 81 studies and report -2.62 as the mean. Pinning price sensitivity to an external elasticity is standard in applied antitrust simulation when good cost-side instruments are missing – the logit merger calibrations of Werden and Froeb (1994) are the template – and it buys transparency: the assumption is one number, open to inspection and to the sensitivity analysis of Appendix C, not something buried in a first stage. Draganska and Jain (2006) estimate structural demand for differentiated CPG product lines and report own-price elasticities in the range used to benchmark the calibrated values here.

2.3 Representation learning and unstructured data in demand estimation

A growing literature swaps observed product characteristics for learned representations. Bajari, Nekipelov, Ryan, and Yang (2015) showed early on that machine-learning methods improve demand prediction. Ruiz, Athey, and Blei (2020) learn latent product representations straight from shopping baskets and use them to answer price counterfactuals; Magnolfi, McClure, and Sorensen (2025) build a product-space embedding from crowd-sourced similarity comparisons and find that embedding distances discipline substitution patterns in ready-to-eat cereal. The shared premise is that the characteristics consumers actually respond to are richer than the handful an econometrician codes by hand, and that data can recover them. The use of unstructured data is most closely related to Compiani, Morozov, and Seiler (2026), who extract embeddings from product images and text with pre-trained deep learning models and incorporate them into a mixed logit, showing across 40 Amazon categories that unstructured data identify close substitutes that attribute-based models miss. This paper shares their premise – embeddings encode the differentiation consumers care about – but differs in implementation and aim. Where Compiani, Morozov, and Seiler (2026) enter embeddings as continuous random-coefficient characteristics in a mixed logit, we use them to *define the* *nesting structure* of a Berry (1994) nested logit. The cost is a coarser substitution pattern; the benefit is a pipeline that stays linear, transparent, and estimable in seconds – which is what the merger-review setting, where speed and auditability come first, actually needs. The embedding technology itself, CLIP, is due to Radford et al. (2021); its contrastive image-text training makes image and text representations directly comparable, which is what lets a single joint product space be built from both modalities.

2.4 Positioning

The contribution sits where the three strands meet. From the demand-and-merger literature it takes the nested-logit inversion and the Bertrand counterfactual; from the calibration literature, an external, inspectable elasticity in place of a fragile first stage; from the representation-learning literature, the idea that a product's place in a learned embedding space is itself an economic characteristic. The new part is the join. The embedding is used not as a regressor but as the *partition* that decides which products compete, and that partition is then put through the placebo tests of Section 5.6 – the internal-validity check that the representation-learning demand papers, focused on fit and counterfactuals, skip. What comes out is a minimal, replicable bridge between modern representation learning and the nested-logit merger-simulation toolkit antitrust practitioners already use.

3 Industry Background and Data

3.1 The hello–Colgate Acquisition

hello Products LLC, founded in 2012, positioned itself as a "naturally friendly" oral-care brand – fluoride-optional formulas, charcoal and coconut-oil lines, playful branding – with strong appeal among younger consumers. Colgate-Palmolive announced its agreement to acquire hello on 23 January 2020 and completed the transaction on 31 January 2020 for $351 million in cash. The acquisition is an attractive empirical laboratory for three reasons: the event date falls cleanly at the start of 2020Q1; the acquirer is the category's largest incumbent while the target occupies a differentiated "natural" niche, so the price effect hinges on exactly the substitution question this paper studies; and both parties sell extensively on Amazon, where listing content is observable.

3.2 Data Sources

The analysis combines two data sources. Household-level purchase histories come from the Open E-commerce 1.0 dataset (Berke et al. 2024), a crowdsourced panel of 5,027 US Amazon consumers with more than 1.8 million transaction records, distributed through Harvard Dataverse. The dataset has been validated by its authors against public Amazon sales data: aggregate expenditure in the panel correlates with published Amazon revenues at Pearson $r = 0.978$ ($p < 0.001$; Berke et al. 2024). The toothpaste purchase records extracted for this study span January 2018 to March 2023 and include order timestamps, prices, and household identifiers; household demographics are available in the source data but enter the analysis only through the count of unique purchasing households per cell. Product-level metadata – titles, benefit descriptions, and image URLs for every purchased ASIN – were retrieved programmatically through the Keepa API (a subscription-based Amazon data service) and parsed with a rule-based extraction script (brand matching against a curated brand list, keyword-based benefit extraction, and unit-aware package-size parsing). Prices are deflated to real terms using the CPI before aggregation; the panel's quarterly price for each ASIN is the mean deflated transaction price across that quarter's purchases.

3.3 Sample Construction

The estimation sample is built in three steps, each dictated by a practical constraint:

1. **Choice-set restriction (167 ASINs).** The universe of toothpaste ASINs is restricted to products with at least five recorded purchases over the sample window. An unrestricted choice set with thousands of rarely purchased alternatives proved computationally infeasible and produced singular design matrices; the five-purchase floor keeps every product with meaningful revealed-preference information while discarding noise listings.

2. **Brand grouping (12 groups).** Brands accounting for more than 2% of purchases are treated as their own groups – Colgate, Crest, SENSODYNE PRONAMEL, Sensodyne, Tom's of Maine, Arm & Hammer, hello, Parodontax, APAGARD, JASON, and Orajel – and all remaining small brands are collapsed into an "Other" category. Together the eleven named groups account for 87% of purchases.

3. **Panel filters (159 ASINs, 1,164 observations).** Purchases are aggregated to ASIN x quarter cells. The estimation window ends in 2022Q3 (later quarters are incompletely covered), and ASINs must appear in at least three quarters to enter the sample, eliminating products whose near-zero shares would be dominated by sampling noise. The resulting panel covers 2018Q1–2022Q3: 19 quarters, 159 ASINs, and 1,164 observations.

Market shares are constructed from revealed-preference purchase counts. Market size M_t is set to $(1+\lambda)\cdot\sum_j q_{jt}$ with $\lambda = 13.93$, implying an outside-good share of approximately 93%.

Separately from these sample filters, the joint embedding space of Section 3.4 is *fitted* on the 155 of 167 ASINs that hold both a valid image and a valid text embedding; the 12 ASINs with text-only embeddings are projected into that space and remain in the estimation sample throughout. Table 3.1 reports summary statistics for the estimation panel.

Table 3.1: Panel Summary Statistics (Estimation Sample)

Variable	Mean	Median	SD	Min	Max
Price (USD)	9.20	8.47	5.34	1.35	41.75
Quarterly units sold	2.17	1.00	1.96	1.00	18.00
Market share (x1,000)	1.093	0.753	1.036	0.335	9.802
Package size (g)	320	259	223	43	1179
Purchasing households per cell	2.06	1.00	1.85	1.00	17.00

Notes:
1,164 ASIN x quarter observations; 159 ASINs; 19 quarters (2018Q1–2022Q3); 12 brand groups. Price is the quarterly mean deflated transaction price per ASIN. Market share uses market size M_t = (1 + 13.93) x total quarterly purchases (outside-good share approx. 93%).

3.4 From Listings to Nests: Embeddings, Joint PCA, and Clustering

Each product j is represented by its listing image and the concatenation of its title and benefit description. CLIP (Contrastive Language-Image Pre-training; Radford et al. 2021) provides an image encoder f_I and a text encoder f_T, trained contrastively so that matching image–text pairs lie close together in a shared representation space. Both encoders (`clip-vit-base-patch32`, a Vision Transformer; Dosovitskiy et al. 2021) map into $\mathbb{R}^{512}$, and outputs are L2-normalised:

$$\mathbf{e}_j^I = \frac{f_I(\text{image}_j)}{\|f_I(\text{image}_j)\|}, \qquad \mathbf{e}_j^T = \frac{f_T(\text{text}_j)}{\|f_T(\text{text}_j)\|}.$$

CLIP's usefulness for this application comes from how it was trained. The model consists of two separate networks: a text Transformer (Vaswani et al. 2017), and – in the variant used here – the ViT-B/32 Vision Transformer, which cuts each image into 32 x 32-pixel patches and processes the patch sequence the way a language model processes tokens (Radford et al. 2021, sec. 2.4; Dosovitskiy et al. 2021). Both networks project into the same 512-dimensional space, and training proceeds contrastively on 400 million (image, caption) pairs collected from the web: within each training batch of N pairs, the model is rewarded for maximising the cosine similarity of the N correct image–caption pairings while minimising the similarity of all mismatched pairings (Radford et al. 2021, sec. 2; their Figure 1 summarises the scheme). Figure 3.1 sketches this objective and its role in the empirical pipeline.

Two properties of the contrastive objective matter here. First, because web captions describe products in ordinary marketing language ("charcoal whitening toothpaste", "fluoride-free kids toothpaste"), the learned space encodes exactly the attribute vocabulary that differentiates consumer goods, with no category-specific training required. Second, because images and text are embedded in one shared space, the two modalities can be combined into a single product representation – which is what the joint PCA below exploits.

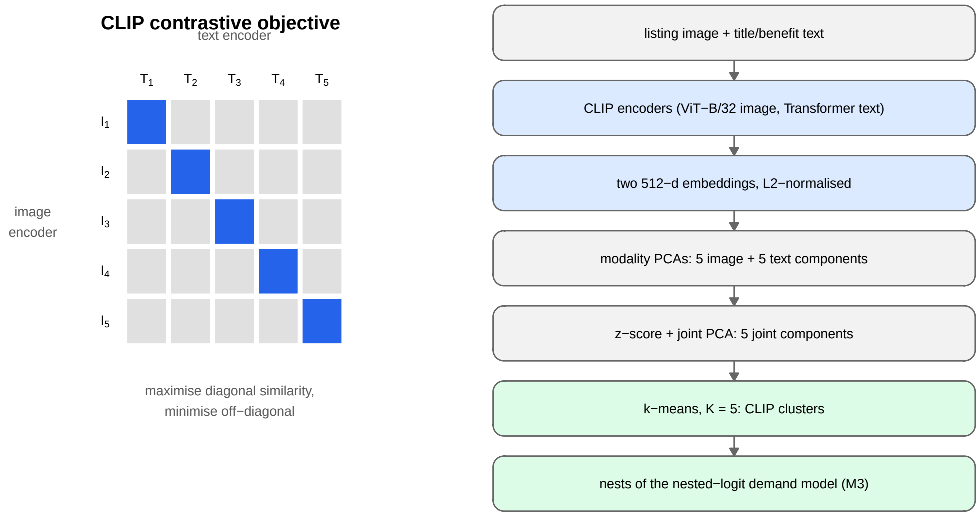

Figure 3.1: How CLIP embeddings are produced and used. Left: the contrastive training objective (after Radford et al. 2021, their Figure 1). An image encoder and a text encoder map a batch of image-caption pairs into a shared space; training pushes the correct pairings (dark diagonal cells) toward high cosine similarity and all mismatched pairings (light cells) apart. Right: the empirical pipeline of this paper, from raw listing content to the nests of model M3.

Each modality's embeddings are first reduced to five modality-specific principal components (Hotelling 1933), with each PCA fitted only on the products holding a valid embedding for that modality. This yields a ten-dimensional characteristics vector per product, $\mathbf{x}_j = (x^I_{j1}, \ldots, x^I_{j5}, x^T_{j1}, \ldots, x^T_{j5})$.

Because these ten columns come from two separate PCA fits, their scales are not comparable: within each modality the first component mechanically carries the largest variance, and the image and text blocks have different overall variance levels. The joint PCA is therefore computed on the correlation scale. Each column k is z-scored using its mean μ_k and standard deviation σ_k over the $N = 155$ products with both modalities,

$$z_{jk} = \frac{x_{jk} - \mu_k}{\sigma_k},$$

and the rotation matrix $\mathbf{W}$ is obtained from the PCA of the standardised matrix $\mathbf{Z}$. The first five joint components, $\mathbf{pc}_j = \mathbf{z}_j' \mathbf{W}_{[\cdot,1:5]}$, define the differentiation space used throughout the paper. Standardisation is not cosmetic: without it, the joint solution would be dominated by whichever input columns happen to carry the largest raw variance – the two first-stage PC1s – rather than by genuine cross-modality structure. The fitted parameters $(\mu_k, \sigma_k, \mathbf{W})$ are saved with the replication materials, which makes the projection of any product into the joint space exactly reproducible.

A useful diagnostic for whether the joint space actually blends the two modalities is the modality share of each component: the fraction of its squared rotation loadings attributable to image inputs,

$$\text{share}^I_m = \frac{\sum_{k \in I} w^2_{km}}{\sum_k w^2_{km}}, \qquad m = 1, \ldots, 5.$$

As Figure 3.2 shows, every retained joint component draws between 48.6% and 51.4% of its loading mass from each modality – the joint space is genuinely multi-modal rather than a relabelled image space or text space.

Finally, nests are formed by k-means clustering (Lloyd 1982) on the five joint components: each product is assigned to one of $K = 5$ clusters,

$$g_j = \arg \min_{g \in \{1,\ldots,K\}} \|\mathbf{pc}_j - \mathbf{c}_g\|^2,$$

with centroids $\mathbf{c}_g$ from 50 random restarts (seed 42), and cluster membership is the nest assignment in M3. The five clusters explain 71.2% of joint-PC variance. Appendix A reports cluster-quality diagnostics – within-cluster sum of squares and average silhouette width (Rousseeuw 1987) – together with all downstream results for $K = 2, \ldots, 10$.

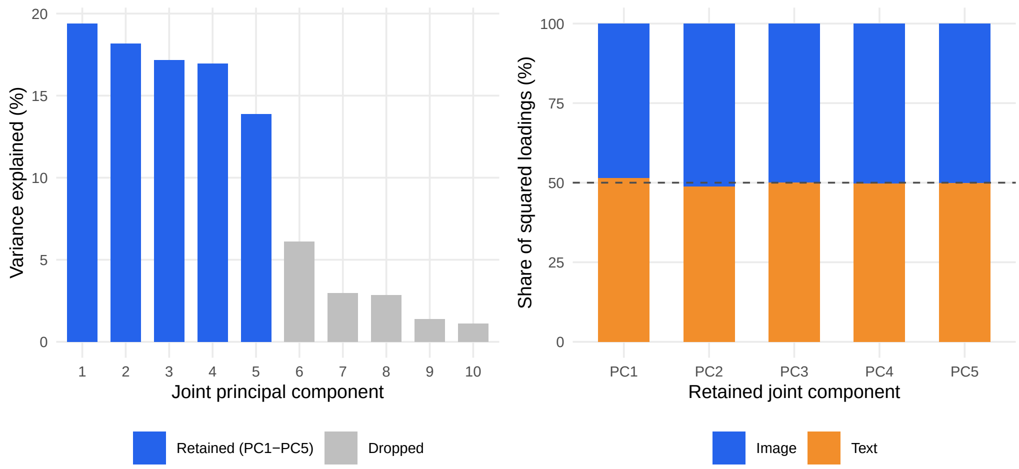

Figure 3.2: Anatomy of the joint embedding space. Left: variance explained by each joint principal component of the standardised ten-column input (five image PCs and five text PCs), fitted on the 155 ASINs with both modalities; the first five components (blue) are retained. Right: modality composition of each retained component, measured as the share of squared rotation loadings attributable to image versus text inputs. The dashed line marks an even 50/50 split; every component lies within 1.4 percentage points of it, confirming the joint space blends both modalities rather than being dominated by either.

Embeddings are static over the sample window. Established Amazon listings change their images and titles only marginally over 2018–2022, so re-extracting embeddings per quarter would produce near-identical vectors; the identifying variation is **cross-sectional** — ASINs within the same brand differ genuinely in packaging and benefit claims — and this is exactly the variation recovered by moving from brand-level to ASIN-level data. Time-varying demand conditions are absorbed by quarter fixed effects.

Twelve ASINs lack an image embedding (image retrieval failed at extraction time) but have valid text embeddings. Rather than dropping them or imputing brand means, they are projected into the joint PC space from their standardised text components, with image dimensions set to the cross-sectional mean – zero in standardised space, so that $\mathbf{pc}_j = (\mathbf{0}, \mathbf{z}_j^T)' \mathbf{W}_{[\cdot,1:5]}$ using the same saved $(\mu_k, \sigma_k, \mathbf{W})$. These ASINs span Crest (4), Sensodyne (2), Tom's of Maine (2), hello (2), and Other (2), and all clear the three-quarter inclusion filter. Dropping them would distort brand-level market shares and remove hello observations inside the merger simulation window; brand-mean imputation was rejected because it would reintroduce precisely the aggregation that the ASIN-level design is meant to avoid.

3.5 The Estimated Product Space

Table 3.2 reports summary statistics by CLIP cluster. The clusters recover economically meaningful segments without using any sales or price information: a premium sensitivity segment (C1, dominated by the GSK brands), the two mainstream incumbents in their own clusters (C2 Colgate, C3 Crest), Tom's of Maine (C4), and a natural/specialty segment (C5) containing hello. Price levels and package sizes differ sharply across clusters, so the clusters capture both vertical and horizontal differentiation.

Table 3.2: Summary Statistics by CLIP Cluster

			Price (USD)			Mkt. Share (x1000)			Pkg. Size (g)		
Cluster	Dom. Brand	N ASINs	$\bar{p}$	$\tilde{p}$	σ_p	$\bar{s} \times 10^3$	$\tilde{s} \times 10^3$	$\sigma_s \times 10^3$	$\bar{w}$	$\tilde{w}$	σ_w
C1	Sensodyne	29	10.26	9.63	5.58	1.068	0.770	0.848	249	249	145
C2	Colgate	33	8.00	8.39	3.65	1.303	0.817	1.259	537	399	296
C3	Crest	29	8.72	6.88	4.18	1.267	0.757	1.282	375	431	128
C4	Tom's of Maine	13	9.85	9.76	2.96	1.170	0.770	0.972	374	370	94
C5	Other (hello)	55	9.46	7.86	6.84	0.839	0.670	0.736	171	141	105

Notes:
Panel: 1,164 ASIN x quarter observations, 19 quarters (2018Q1–2022Q3). Mean/Median/SD of price in USD, market share scaled x1000, package weight in grams. Dominant brand = modal brand by ASIN count within cluster.

Beyond price and share statistics, the clusters are directly interpretable from the listing content itself. Table 3.3 reports, for each cluster, the listing-vocabulary terms that are most over-represented relative to the whole category (by lift: the share of the cluster's ASINs whose title or benefit text contains the term, divided by the category-wide share). The vocabulary confirms that the clusters track product positioning, not arbitrary embedding geometry.

Table 3.3: Distinctive Listing Vocabulary by CLIP Cluster

Cluster	Dominant brand	N ASINs	Most over-represented listing terms (by lift)
C1	Sensodyne	29	sensodyne, parodontax, extra, bleeding, pronamel, reharden
C2	Colgate	33	optic, hydrogen, max, benefits, extreme, frosty
C3	Crest	29	scope, pro, radiant, brilliance, eraser, glamorous
C4	Tom's of Maine	13	maine, tom's, children's, silly, animals, not
C5	Other	55	sls, activated, mouth, orajel, remineralizing, anti

Notes:
Terms from ASIN titles and benefit descriptions (lower-cased, unit and category stopwords removed). Lift = share of the cluster's ASINs containing the term divided by the category-wide share; terms shown appear in at least two of the cluster's ASINs and have lift above 1.5, ranked by lift. Cluster assignment identical to the estimation (K = 5, seed 42).

Figure 3.3 visualises the CLIP product space: hello's listings sit in the natural/specialty cluster on the left of the embedding plane while Colgate's cluster occupies the opposite region, making the low implied diversion between the merging parties directly visible.

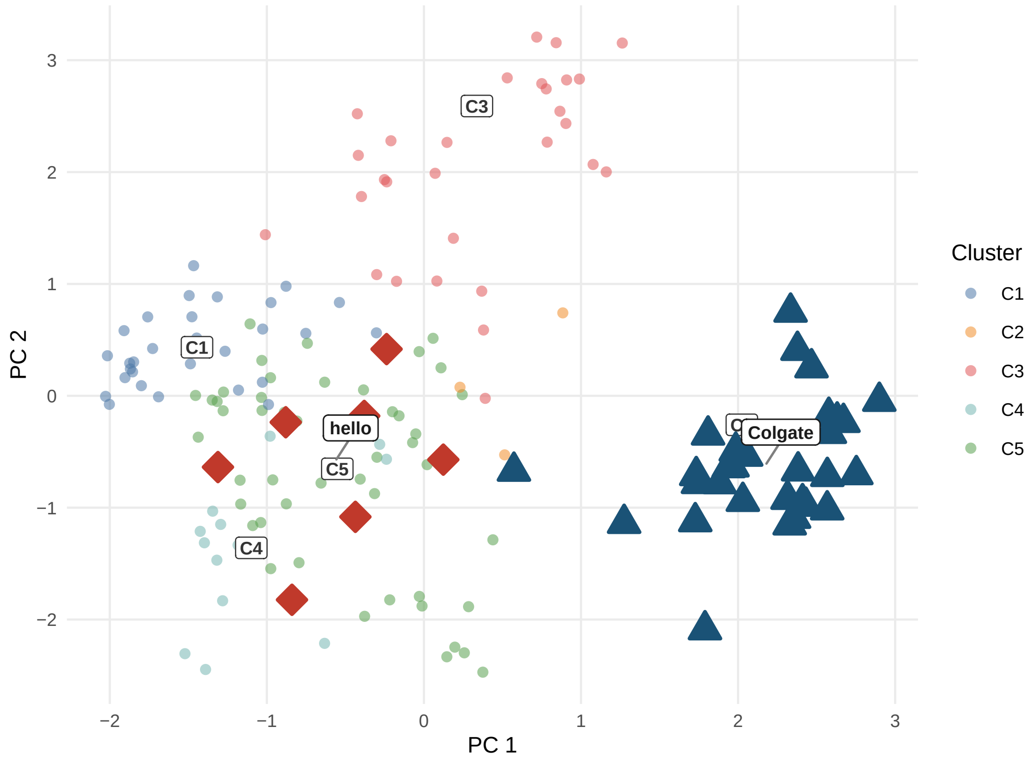

Figure 3.3: Product space from CLIP embeddings. Each point is one ASIN positioned by its first two principal components of the joint image+text embedding. Cluster centroids are labelled C1–C5. hello (red diamonds, C5) and Colgate (dark blue triangles, C2) occupy different clusters, implying a lower diversion ratio and a smaller predicted merger effect under CLIP nesting relative to plain logit.

4 Empirical Framework

4.1 Berry (1994) Linearisation

All three models are estimated via the Berry (1994) nested-logit linearisation:

$$\log \frac{s_{jt}}{s_{0t}} - \rho \log s_{j|g,t} = \alpha\, p_{jt} + \boldsymbol{\beta}'\mathbf{x}_{jt} + \mathrm{FE}_t + \xi_{jt}$$

where $s_{j|g,t} = s_{jt}/\sum_{k\in g} s_{kt}$ is the within-nest share, $\rho \in [0,1)$ is the nesting intensity, and FE_t are quarter fixed effects (absorbed by within-quarter demeaning). The underlying choice model is the generalised-extreme-value family of McFadden (1974, 1978), of which the nested logit is the leading closed-form member; the error-components representation that motivates ρ is due to Cardell (1997), and Train (2009, chs. 3–4) gives a textbook treatment.

The estimating equation is derived, not assumed, and the single step that matters for this paper – the one where its contribution enters – is worth walking through. Consumer i's utility from product j in nest g is $u_{ijt} = \delta_{jt} + \zeta_{igt} + (1-\rho)\,\varepsilon_{ijt}$, with mean utility $\delta_{jt} = \alpha\, p_{jt} + \boldsymbol{\beta}'\mathbf{x}_{jt} + \mathrm{FE}_t + \xi_{jt}$, an i.i.d. extreme-value ε_{ijt}, and the nest-specific term ζ_{igt} of Cardell (1997) that ties products within a nest together with strength ρ. Let $D_{gt} = \sum_{k\in g} \exp\bigl(\delta_{kt}/(1-\rho)\bigr)$ be the nest's inclusive value. The GEV choice probabilities then split cleanly into a within-nest piece and a between-nest piece,

$$s_{jt} = s_{j|g,t}\cdot s_{g,t}, \qquad s_{j|g,t} = \frac{\exp\bigl(\delta_{jt}/(1-\rho)\bigr)}{D_{gt}}, \qquad s_{g,t} = \frac{D_{gt}^{1-\rho}}{\sum_{g'} D_{g't}^{1-\rho}}.$$

Treat the outside good as a singleton nest with $\delta_{0t} = 0$. Then $s_{g,t}/s_{0t} = D_{gt}^{1-\rho}$, and $\log s_{jt} - \log s_{0t} = \log s_{j|g,t} + (1-\rho)\log D_{gt}$. Rearrange the within-nest share to $\log D_{gt} = \delta_{jt}/(1-\rho) - \log s_{j|g,t}$, substitute, cancel, and the Berry (1994) linearisation falls out:

$$\log \frac{s_{jt}}{s_{0t}} = \delta_{jt} + \rho \log s_{j|g,t}.$$

So ρ reaches the regression through exactly one observable term, $\log s_{j|g,t}$, and the nest g shows up *only* in how that within-nest share is composed. Nothing in the equation dictates what a nest must be – any partition of the products into groups yields a valid $s_{j|g,t}$. That is the opening this paper works through: Section 3.4 builds g from CLIP embedding clusters instead of brand identity, and leaves every other term untouched.

4.2 Price Coefficient Calibration

Valid cost-side instruments for price are unavailable in this high-frequency online retail dataset (see Section 5). Following the recommendation of Berry (1994) and standard practice in applied antitrust simulation, α is calibrated to match a target median own-price elasticity:

$$\alpha = \frac{\varepsilon_{\text{target}}}{\text{median}(p_{jt}(1 - s_{jt}))} = \frac{-2.62}{8.4502} = -0.310$$

The target $\varepsilon_{\text{target}} = -2.62$ corresponds to the brand-level CPG meta-analysis mean of Bijmolt, van Heerde & Pieters (2005, p. 141), consistent with structural BLP estimates for analogous packaged-goods categories (Nevo 2001; Draganska & Jain 2006).

Sequential calibration note. Since α is calibrated at $\rho = 0$, the target elasticity -2.62 is the plain-logit median. At the profiled nesting parameters, the nested-logit elasticity formula amplifies own-price sensitivity by roughly $1/(1 - \rho^*)$, so the same α implies median own-price elasticities of -3.83 under M2 and -7.31 under M3 (computed observation by observation; the $1/(1 - \rho^*)$ back-of-envelope gives -4.0 and -7.9). These larger magnitudes are not in tension with the calibration target. The Bijmolt, van Heerde, and Pieters (2005) mean is dominated by brand- and aggregate-level estimates, whereas the products here are individual ASINs, which face much closer substitutes than a brand aggregate does – finer product definitions mechanically imply more elastic demand. Structural estimates at the product-line level are correspondingly larger: Draganska and Jain (2006) report own-price elasticities between -2.45 and -6.25, and the implied medians here sit at the top of and beyond that range (-3.83 under M2, -7.31 under M3) – exactly the pattern the finer ASIN-level product definition predicts, since individual listings face closer substitutes than the product lines in their data. Appendix C reports a robustness exercise that rescales α within each nested model so that the model's own implied median equals -2.62 at its profiled ρ^*: doing so shrinks $|\alpha|$, inflates Bertrand markups, and drives the negative-cost rate far above even the plain-logit level. In other words, ASIN-level demand must be substantially more elastic than the aggregate benchmark for observed prices to be rationalizable at all – the cost side independently validates the more elastic implied elasticities.

4.3 ρ^* Selection — Concentrated OLS Profile

For each ρ on a fine grid $\{0, 0.01, \ldots, 0.90\}$, the adjusted dependent variable is formed and $\boldsymbol{\beta}(\rho)$ is estimated by OLS. The profiled ρ^* minimises the residual sum of squares:

$$\tilde{y}_{jt}(\rho) = \log \frac{s_{jt}}{s_{0t}} - \alpha p_{jt} - \rho \log s_{j|g,t}, \qquad \rho^* = \arg\min_{\rho} \left\| \tilde{y}(\rho) - \mathbf{X}\hat{\boldsymbol{\beta}}(\rho) \right\|^2$$

4.4 Marginal Costs and Merger Simulation

Marginal costs are recovered from the pre-merger Bertrand first-order condition (Berry 1994):

$$\mathbf{mc} = \mathbf{p} + (\Omega_{\text{pre}} \circ \Delta)^{-1}\mathbf{s}$$

where $\Omega[j,k] = \mathbf{1}[\text{firm}_j = \text{firm}_k]$ and Δ is the demand Jacobian with elements:

$$\Delta_{jj} = \alpha s_j\left[\tfrac{1}{1-\rho} - \tfrac{\rho}{1-\rho}s_{j|g} - s_j\right], \quad \Delta_{jk}^{\text{same}} = \alpha s_j\left[-\tfrac{\rho}{1-\rho}s_{k|g} - s_k\right], \quad \Delta_{jk}^{\text{diff}} = -\alpha s_j s_k$$

Remark (brand nests and ownership). Marginal-cost recovery interacts with the nest assignment in one important, exact way. When every nest lies wholly inside one firm – as with brand nests, where each firm owns complete brands – the nested-logit Bertrand first-order conditions admit the same solution as plain logit: markups are uniform within each firm at $1/(|\alpha|(1 - S_f))$, where S_f is the firm's inside share, for *any* value of ρ (proof in Appendix D). Consequently, M1 and M2 recover *identical* marginal costs at every observation, identical Lerner indices, and identical negative-cost rates. Brand nesting is therefore incapable of repairing the cost side of the model in this market by construction; only a nest assignment that cuts across ownership boundaries – such as the CLIP clusters, which group ASINs of different firms – can change recovered costs. The counterfactual equilibria of M1 and M2 still differ slightly, because the share *functions* $s(p)$ differ even though markups at observed shares coincide.

Marginal costs are averaged per ASIN across pre-merger quarters and held fixed in both counterfactuals. Holding costs at pre-merger values serves two purposes: it embodies the standard no-efficiencies assumption of prospective merger review (Farrell and Shapiro 1990), and it avoids a mechanical tautology – if costs were re-inverted in post-merger quarters, the no-merger Nash equilibrium would recover observed prices by construction, since the first-order condition is the identity that defined those costs.

Two Nash equilibria are then computed for every post-merger quarter via a damped fixed-point iteration on the Bertrand first-order conditions (Morrow and Skerlos 2011):

$$\mathbf{p}_{n+1} = \lambda\left[\mathbf{mc} - (\Omega \circ \Delta(\mathbf{p}_n))^{-1}\mathbf{s}(\mathbf{p}_n)\right] + (1-\lambda)\,\mathbf{p}_n$$

with $\lambda = 0.4$, convergence tolerance 10^{-8}, and up to 2,000 iterations. The no-merger counterfactual uses pre-merger ownership Ω_{pre}; the merger simulation replaces it with Ω_{post} (hello ASINs assigned to Colgate's firm identifier from 2020Q1). The merger price effect is the difference between the two equilibria, $\Delta p = p^{\text{merger}} - p^{\text{no-merger}}$, so that ownership structure is the only thing that varies between them.

Algorithm validation. The R implementation mirrors the counterfactual equilibrium routines in PyBLP (Conlon & Gortmaker 2020) and is validated against

PyBLP's `compute_prices()` directly: for both M1 (plain logit) and M3 (CLIP-nested, $\rho^* = 0.67$), the maximum absolute price difference across 847 post-merger observations is below 2×10^{-8}, confirming algorithmic equivalence.

5 Results

5.1 Demand Estimates

Table 5.1 presents the demand parameters for all three models, separating the calibrated α from the estimated $\boldsymbol{\beta}$ and the profiled ρ^*. Brand nesting yields $\rho^* = 0.35$ with a 3.1% RSS improvement over plain logit; CLIP nesting yields $\rho^* = 0.67$ with an 8.1% improvement. The stronger profiled nesting parameter indicates that within-cluster substitution is substantially more intense when clusters are defined in embedding space than when they are defined by brand identity.

Table 5.1: Demand Parameter Estimates — Three Models

Variable	M1: Logit (ρ=0)	M2: Brand-Nested (ρ^*=0.35)	M3: CLIP-Nested (ρ^*=0.67)
α (price)	**$-$0.310 (a)**	**$-$0.310 (a)**	**$-$0.310 (a)**
Package size (g)	0.0037*** (0.0003)	0.0037*** (0.0003)	0.0034*** (0.0003)
CLIP PC_1	-0.4034*** (0.0424)	-0.2869*** (0.0418)	-0.3280*** (0.0407)
CLIP PC_2	0.2033*** (0.0374)	0.2681*** (0.0368)	0.2103*** (0.0359)
CLIP PC_3	-0.1667*** (0.0386)	-0.1479*** (0.0380)	-0.2854*** (0.0370)
CLIP PC_4	0.2115*** (0.0399)	0.1574*** (0.0393)	0.2707*** (0.0382)
CLIP PC_5	-0.1652*** (0.0390)	-0.1243*** (0.0384)	-0.1745*** (0.0374)
Other brand	0.0849 (0.1327)	0.4590*** (0.1307)	0.2641** (0.1272)
Import x freight shock	0.6767*** (0.2342)	0.6709*** (0.2306)	0.7461*** (0.2245)
ρ^*	0.000 (b)	0.350 (b)	0.670 (b)
Quarter FEs	Yes	Yes	Yes
Median implied ε_{jj}	-2.62	-3.83	-7.31
RSS drop vs ρ=0	—	3.1%	8.1%

Notes:
*** p<0.01, ** p<0.05, * p<0.10. FEs by quarter. N=1,164. (a) calibrated alpha; (b) grid-searched rho*. Sections 4.2-4.3.

5.2 Own-Price Elasticities by Brand

Table 5.2 reports median own-price elasticities by brand group for each model. Two patterns are worth noting. First, within each model, elasticities scale with price levels. Because the nested-logit own-price elasticity is proportional to αp_j, premium brands such as APAGARD and SENSODYNE PRONAMEL are mechanically the most elastic. Second, and more informative, moving from M1 to M3 amplifies all elasticities by roughly $1/(1-\rho^*)$, but the amplification interacts with cluster composition: brands sharing a crowded embedding cluster face the strongest within-nest competition. The merging parties show the pattern. hello's median elasticity rises from -1.42 under plain logit to -4.05 under CLIP nesting, because its close substitutes are the natural/specialty products in its own cluster, not Colgate.

Table 5.2: Median Own-Price Elasticities by Brand Group

Brand	N obs.	M1: Logit	M2: Brand-Nested	M3: CLIP-Nested
APAGARD	44	-4.92	-6.62	-14.33
Sensodyne	99	-3.00	-4.27	-8.68
SENSODYNE PRONAMEL	90	-3.04	-4.42	-8.67
Colgate	**224**	**-2.84**	**-4.27**	**-8.24**
Tom's of Maine	85	-3.03	-4.38	-8.07
Other	220	-2.70	-4.09	-7.98
Crest	219	-2.11	-3.18	-6.17
Parodontax	51	-1.79	-2.54	-5.29
Arm & Hammer	21	-1.53	-1.62	-4.22
hello	**50**	**-1.42**	**-1.88**	**-4.05**
JASON	29	-1.30	-1.48	-3.77
Orajel	32	-0.86	-1.16	-2.53

Notes:
Median own-price elasticity across all ASIN x quarter observations of each brand group, full estimation panel (1,164 observations). Elasticities use the nested-logit formula of Section 4.4 with the calibrated alpha (-0.310) and each model's profiled rho (0 for M1, 0.35 for M2, 0.67 for M3). Rows sorted by M3 elasticity; bold = merger parties.

5.3 Diversion Ratios

The merger question turns on where hello's demand goes when its prices rise. Table 5.3 reports diversion ratios $DR_{j\to k} = -(\partial s_k/\partial p_j)/(\partial s_j/\partial p_j)$ from hello ASINs, aggregated into fixed reporting buckets (share-weighted across hello products and pre-merger quarters). Diversion ratios are the central input of modern unilateral-effects screening (Farrell and Shapiro 2010; Conlon and Mortimer 2021), and they enter the U.S. Horizontal Merger Guidelines directly through the pricing-pressure analysis of differentiated products (U.S. Department of Justice and Federal Trade Commission 2010, sec. 6.1).

Table 5.3: Diversion from hello by Destination (percent of lost demand)

Model	To Colgate	To rest of hello's CLIP cluster	To other inside goods	To outside good
M1: Logit	1.16	1.49	3.88	93.47
M2: Brand-Nested	0.93	1.20	22.41	75.45
M3: CLIP-Nested	**0.41**	**58.14**	**8.39**	**33.06**

Notes:
Share-weighted average across hello ASINs and pre-merger quarters (2018Q1–2019Q4), computed from each model's demand Jacobian at observed shares with the calibrated alpha and the model's profiled rho*. Reporting buckets are fixed across models: Colgate-brand products, remaining products in hello's CLIP cluster (C5), all other inside goods, and the outside good. Rows may not sum to 100 due to rounding.

The three demand systems tell three different stories about the same market. Under plain logit, diversion is proportional to market shares, so 93% of hello's marginal demand flows to the outside good and only 1.2% reaches Colgate. Brand nesting redirects substitution toward hello's own sister ASINs but leaves cross-brand diversion share-proportional. Under CLIP nesting, 58% of hello's lost demand diverts to the other natural/specialty products in its cluster, while diversion to Colgate falls to 0.4% – the low-diversion configuration that drives the small simulated merger effect in the next subsection.

5.4 Merger Counterfactuals

Table 5.4 reports the merger counterfactuals: observed prices, recovered marginal costs, the no-merger Nash equilibrium, the merger Nash equilibrium, and the implied price effect Δp for the merging parties and the largest rivals. Under plain logit, where diversion is proportional to market shares, the simulated effect for hello is +1.23%; brand nesting barely changes it (+1.28%) because hello's ASINs sit in their own brand nest and diversion to Colgate is still share-proportional across nests. Under CLIP nesting, hello (cluster C5) and Colgate (cluster C2) are distant in product space, diversion between the parties is small, and the simulated hello effect falls to +0.40%, with Colgate's own effect economically negligible.

Table 5.4: Merger Counterfactuals: hello-to-Colgate Acquisition (2020Q1)

Brand	Obs. p	MC	Nash (pre)	Nash (post)	Δp	Δp %
M1: Logit (ρ=0)						
hello	**6.43**	**3.20**	**7.10**	**7.16**	**0.064**	**1.233%**
Colgate	**8.45**	**5.16**	**8.29**	**8.30**	**0.007**	**0.104%**
Tom's of Maine	10.47	7.17	10.70	10.71	0.006	0.069%
Crest	8.87	5.60	9.18	9.18	0.000	0.000%
Sensodyne	10.45	7.17	10.05	10.05	0.000	0.000%
M2: Brand-Nested (ρ^*=0.35)						
hello	**6.43**	**3.20**	**7.10**	**7.17**	**0.065**	**1.275%**
Colgate	**8.45**	**5.16**	**8.30**	**8.30**	**0.006**	**0.104%**
Tom's of Maine	10.47	7.17	10.71	10.71	0.006	0.069%
Crest	8.87	5.60	9.18	9.18	0.000	0.000%
Sensodyne	10.45	7.17	10.07	10.07	0.000	0.000%
M3: CLIP-Nested (ρ^*=0.67)						
hello	**6.43**	**5.20**	**7.05**	**7.07**	**0.020**	**0.404%**
Colgate	**8.45**	**5.88**	**8.59**	**8.59**	**0.002**	**0.018%**
Tom's of Maine	10.47	7.19	10.72	10.72	0.002	0.021%
Crest	8.87	5.63	9.18	9.18	0.000	0.000%
Sensodyne	10.45	7.28	10.29	10.29	0.000	0.000%

Notes:
All prices USD; post-merger averages (2020Q1–2022Q3). Bold = merger parties. Nash (pre): Bertrand-Nash under pre-merger ownership. Nash (post): equilibrium under merged ownership. Dp = Nash (post) minus Nash (pre). MC inverted from pre-merger FOC and held fixed.

Figure 5.1 traces the post-merger window quarter by quarter, showing the merger Nash equilibrium diverging from the no-merger counterfactual for hello while the two paths for Colgate remain indistinguishable.

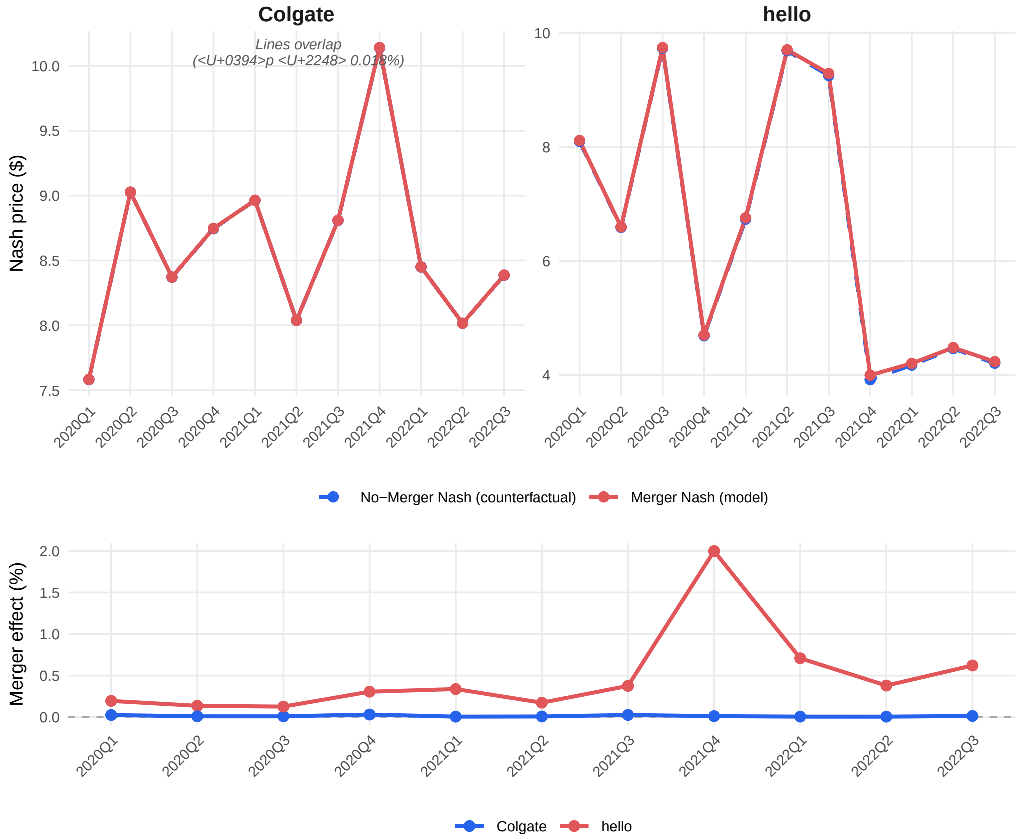

Figure 5.1: Merger simulation price trajectories (Model 3, CLIP-Nested, $\rho^* = 0.67$). Top panel: Nash equilibrium prices under merged and pre-merger ownership. Observed prices are omitted from the top panel because their quarter-to-quarter variation ($4–$10) would compress the counterfactual divergence. Bottom panel: merger price effect in percent, $(p^{\text{merger}} - p^{\text{no-merger}})/p^{\text{no-merger}} \times 100$. hello (red) shows a positive and time-varying effect; Colgate (blue) remains near zero.

5.5 Cross-Model Comparison

The central economic-plausibility comparison across the three models is summarised below. The share of observations with **negative implied marginal costs** is the key diagnostic: when the demand system understates substitutability, the Bertrand inversion attributes implausibly large markups, pushing recovered costs below zero. CLIP-based nesting cuts the negative-MC rate from 8.6% to 1.7%, a five-fold reduction, and at the same time gives the largest RSS gain. That M1 and M2 display *exactly* the same negative-cost rate, cost levels, and Lerner indices is not a coincidence: because brand nests coincide with ownership boundaries, brand-nested Bertrand markups are identical to plain-logit markups for any ρ (Section 4.4 and Appendix D). The cost side of the model can only improve when nests cut across firms, which is precisely what CLIP clustering delivers. The final column translates the price effects into consumer terms via the nested-logit inclusive-value formula (Small and Rosen 1981); implied consumer harm under CLIP nesting is roughly a third of the logit-implied harm.

Table 5.5: Cross-Model Comparison: Demand Fit and Economic Plausibility

Model	ρ^*	Med. ε_{jj}	RSS drop	Neg. MC %	Med. Lerner	hello Δp	Colgate Δp	ΔCS
M1: Logit	0.00	-2.62	—	8.6%	38.5%	+1.23%	+0.104%	-25.4
M2: Brand-Nested	0.35	-3.83	3.1%	8.6%	38.5%	+1.27%	+0.104%	-26.6
M3: CLIP-Nested	**0.67**	**-7.31**	**8.1%**	**1.7%**	**28.9%**	**+0.40%**	**+0.018%**	**-8.5**

Notes:
All statistics over the full estimation panel (1,164 ASIN x quarter observations) except merger effects (post-merg quarters 2020Q1–2022Q3). Med. elasticity = median own-price elasticity. RSS drop = reduction in residual su of squares vs rho=0 from the concentrated OLS profile. Neg. MC = share of observations with negative implie marginal cost from the Bertrand inversion (lower = more economically plausible). Med. Lerner = median (p - mc)/ hello / Colgate Dp = average post-merger Nash price effect. dCS = mean change in consumer surplus, merger no-merger Nash, in cents per 1,000 consumers per quarter (nested-logit log-sum formula). Bold/green row = preferre specification.

5.6 Placebo Nesting Tests

The negative-MC result invites a sharp question: is the improvement specific to the *semantic content* of the CLIP embeddings, or would any five-way partition that cuts across ownership boundaries do just as well? Appendix D says the concern is real – the cost side can only improve when nests cross firms. So we test it head-on. Two placebo nestings, 200 draws each, go through the identical pipeline of calibration, ρ profile, cost inversion, and merger simulation: (A) every ASIN assigned to one of five nests uniformly at random, and (B) the CLIP cluster labels permuted across ASINs, which keeps the CLIP nest-size distribution but severs the product-to-embedding link. Before any draw counts, the harness must reproduce the real CLIP figures exactly (hello effect +0.404%); the placebos are measured on the same ruler.

Table 5.6 and Figure 5.2 split what is generic from what is CLIP's own.

The cost-side repair is generic. Random cross-firm nests push the negative-MC rate down to a median of 0.9%; they match or better CLIP's 1.7% in 96% of draws. This is what Appendix D predicts – crossing ownership is what does the work. The 8.6%-to-1.7% drop is real and it matters, but it cannot, on its own, tell CLIP apart from noise.

The demand fit is not generic. CLIP's RSS drop of 8.1% beats 97.5% of the random partitions (their median: 5.3%), and its $\rho^* = 0.67$ beats 97% of them. A nesting only lowers residual variance when the products grouped together actually share demand shocks. Embedding-similar products do; randomly grouped products mostly add noise.

The merger prediction is not generic – and this is the decisive result. Every one of the 200 random draws puts hello's price effect above CLIP's +0.40%; the draws range from +2.9% at the minimum to +10.3% at the maximum (median +5.9%, a fourteen-fold gap; Table 5.6 reports the narrower 5–95% band). The mechanism is economic, not statistical. Colgate sells many ASINs. Scatter them randomly over five nests and hello's nest almost always catches a few, manufacturing hello-to-Colgate diversion with no economic basis behind it – and the simulated effect inflates accordingly. CLIP keeps Colgate's entire product line in one cluster, cleanly apart from hello's natural/specialty cluster, so whatever diversion remains between the merging parties reflects actual product differentiation. Placebo B sharpens the point: permuting the CLIP labels holds nest sizes exactly fixed, yet the negative-MC rate still falls (median 1.6%) while the hello effect still inflates (median +5.2%). What CLIP gets right is the *assignment* of products to nests, not their sizes.

Nor is a hand-coded classification a substitute. Could a human analyst reading the listings simply build these nests by hand? We tried. A keyword classifier on the raw title and benefit text (kids, sensitivity, whitening, natural, standard), run in two tie-breaking orders because most listings carry several benefit claims, goes through the same pipeline as everything else. The keyword nests fit demand worse than the *median random partition* – RSS drops of 4.2% and 4.4% against the random median of 5.3% – and both variants put hello's price effect above +4.5%. The natural-first variant fails in the most instructive way. It does separate hello from the Colgate brand; it also pulls thirteen Tom's of Maine listings, Colgate-owned, into hello's nest, because Tom's carries the same natural keywords. CLIP assigns hello, Colgate, and Tom's of Maine to three *different* clusters. The embeddings register that hello's bright, design-led positioning is not Tom's of Maine's earth-toned health-store positioning – a distinction no keyword list can encode – and the demand data reward exactly that distinction with the largest fit improvement.

Together, the three results give the precise statement of the contribution. Any cross-firm nesting can repair the cost side. Only CLIP repairs it *while* fitting demand better than 97.5% of random partitions and delivering a merger counterfactual free of the spurious diversion those partitions introduce. Brand nesting sits at the other extreme – it separates the merging parties but leaves the cost side untouched (Appendix D) – and a keyword classification fares no better, fitting worse than the median random draw while co-nesting hello with Colgate's natural subsidiary. Of the nesting rules considered, CLIP alone satisfies all three at once.

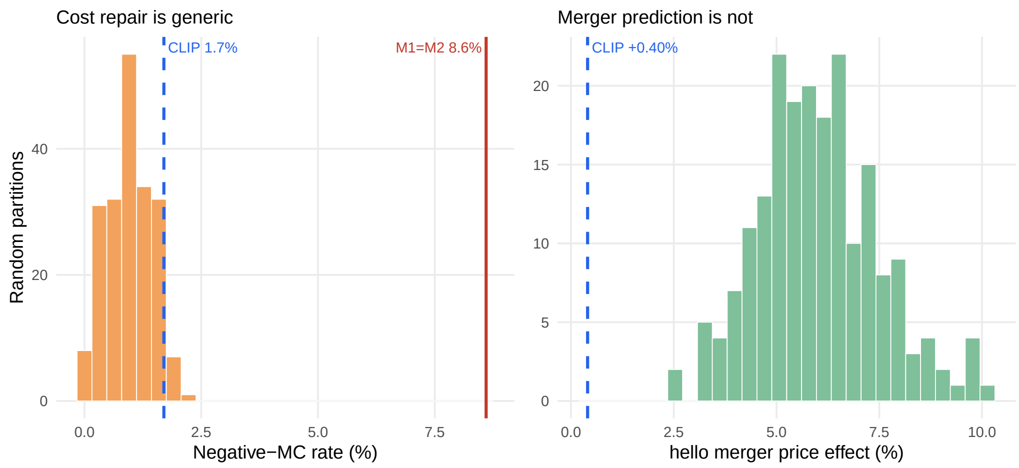

Figure 5.2: Placebo nesting tests (200 random cross-firm partitions). Left: the negative-MC rate under random nests (bars) reaches CLIP's level (blue dashed) or lower in 96% of draws – the cost-side repair is generic, far below the M1=M2 baseline (red). Right: the simulated hello merger price effect under random nests ranges from +2.9% to +10.3%; CLIP's +0.40% (blue dashed) lies below all 200 draws, because CLIP alone keeps Colgate's product line out of hello's nest.

Table 5.6: Placebo and Baseline Nesting Tests: CLIP vs Random, Shuffled, and Hand-Coded Partitions

Metric	CLIP (real)	Random A (median [5–95%])	Shuffled B (median)	Keywords (nat.-first)
ρ^*	0.67	0.53 [0.39, 0.66]	0.41	0.44
RSS drop (%)	8.10	5.30 [3.00, 7.91]	4.00	4.40
Neg-MC (%)	1.70	0.90 [0.20, 1.70]	1.60	1.90
hello effect (%)	**0.40**	**5.87 [3.80, 8.73]**	**5.15**	**5.52**
Colgate effect (%)	0.02	0.41 [0.26, 0.59]	0.39	0.08

Notes:
200 draws each. Random A: every ASIN assigned to one of five nests uniformly at random (cross-firm, no semantics). Shuffled B: CLIP cluster labels permuted across ASINs (CLIP nest-size distribution preserved). All draws run through the identical calibration, rho profile, cost inversion, and merger simulation as M3. Keywords = hand-coded keyword categories from the raw listing text, natural-first tie-breaking (the benefit-first variant is similar: RSS drop 4.2%, hello +4.51%). hello effect = average post-merger Nash price effect for hello. Bold row = the decisive result: CLIP's +0.40% lies below all 200 random draws (min +2.90%).

6 Discussion and Limitations

The findings should be read with the design's limitations in view; each is a deliberate trade-off rather than an oversight, and each points to a concrete extension.

Calibrated price coefficient. Valid cost-side instruments are unavailable in this high-frequency online retail panel, so α is calibrated to an external meta-analytic elasticity rather than estimated. The choice makes the price-sensitivity assumption transparent, but it also means α is common across products and the level of all markups inherits the calibration. Because α is calibrated at $\rho = 0$, the nested models imply more elastic demand at their profiled ρ^*; the cross-model *comparison* of fit, negative-cost rates, and merger effects is unaffected, since all three models share the same α. A natural extension is to construct differentiation instruments (Gandhi and Houde 2019) from the CLIP embeddings themselves – rival products' embedding distances are plausibly excluded from a product's own demand shock – and estimate α rather than calibrate it.

Demand-side cost recovery. Marginal costs are recovered solely from the demand system and the Bertrand first-order conditions; without a supply-side cost equation, true cost variation cannot be separated from demand shocks absorbed into the inversion. The negative-cost rate is used here precisely as a *diagnostic* of demand misspecification, which is robust to this concern, but the recovered cost levels should not be interpreted structurally.

Fixed marginal costs and no efficiencies. The counterfactuals hold costs at pre-merger levels, isolating the market-power effect of the ownership change. If the acquisition generated real marginal-cost efficiencies, the simulated price effects are upper bounds for the merged firm's products.

Static embeddings and cluster count. Embeddings are extracted once per product, which is appropriate for listings that change little, but would miss repositioning. The number of clusters is selected at $K = 5$; Appendix A shows the qualitative conclusions – high ρ^*, low negative-cost rates, small hello effects – are stable across $K = 2$ to 10, though point estimates move.

Sparse purchase counts. The crowdsourced panel records 2,787 toothpaste purchases across 1,164 ASIN x quarter cells – a median of one purchasing household per cell – so individual shares carry sampling noise. The three-quarter presence filter, quarterly aggregation, and the focus on cross-model *comparisons* (which hold the data fixed) mitigate this, but individual cell-level quantities should be read with appropriate caution.

External validity. The analysis covers one category on one platform around one merger. The pipeline, however, requires only listing content, purchase counts, and prices – data available for essentially any e-commerce category – which is what makes the approach portable to other merger settings.

7 Conclusion

This paper provides evidence that substitution patterns constructed from unstructured product content outperform those built from brand labels in a standard merger-simulation framework. Replacing brand nests with CLIP-embedding clusters in a Berry (1994) nested logit raises demand fit, cuts the rate of economically implausible negative implied marginal costs from 8.6% to 1.7%, and changes the antitrust answer: the simulated price effect of the hello–Colgate acquisition for hello falls from +1.23% under plain logit to +0.40% under CLIP nesting, because the embeddings reveal that the merging parties occupy different regions of product space and divert little demand to one another.

The broader message for practice is that the inputs to credible merger simulation need not be limited to structured characteristics data. Product images and listing text, observable for any online market at essentially zero cost, carry the differentiation consumers act on, and a pre-trained encoder plus standard clustering suffices to put that information into the tools antitrust economists already use. The entire analysis, from embeddings to Nash counterfactuals, is reproducible from a single document and a public repository, which is intended to make the method auditable in exactly the settings where it would be applied.

Data and Code Availability

All data, code, and the R Markdown source of this paper are openly available at `https://github.com/brkuzn/clip-amazon-toothpaste-market`. Knitting the source document re-runs the entire pipeline – the joint PCA, the ρ grid searches, and the Bertrand–Nash equilibria – and regenerates every table and figure from the included data. The repository also contains the CLIP extraction notebook, the joint-PCA construction script, and reference copies of all outputs; reproduction of the shipped joint principal components is byte-exact. Raw household-level purchase records with demographics are not redistributed and are available from the Open E-commerce 1.0 repository (Berke et al. 2024).

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Appendix

Appendix A: K Sensitivity Analysis

Table 7.1: K Sensitivity Analysis: CLIP Cluster Count K = 2–10

K	WCSS	Silhouette	ρ^*	RSS Drop	Neg. MC%	hello Δp%	Colgate Δp%
2	1042.0	0.199	0.60	6.1%	0.5%	1.71%	0.015%
3	836.7	0.256	0.66	7.8%	0.9%	3.49%	0.013%
4	641.2	0.330	0.57	5.7%	3.0%	3.96%	0.041%
5	**456.5**	**0.388**	**0.67**	**8.1%**	**1.7%**	**0.40%**	**0.018%**
6	365.6	0.387	0.60	6.7%	3.4%	0.59%	0.044%
7	322.0	0.368	0.48	4.6%	4.4%	0.69%	0.055%
8	279.3	0.400	0.37	2.9%	4.7%	0.82%	0.067%
9	247.7	0.403	0.35	2.4%	4.7%	1.26%	0.177%
10	230.7	0.373	0.40	3.8%	4.6%	0.79%	0.065%

Notes: K-means on 5 CLIP PCs (seed 42, 50 restarts). WCSS = within-cluster sum of squares. Silhouette = average silhouette width (higher = better separation). ρ^* = argmin RSS on grid [0, 0.01, . . . , 0.90]. RSS drop = $100 \times (RSS_0 - RSS^*)/RSS_0$. Neg. MC = share of ASINs with negative implied marginal cost. hello / Colgate Δp = average post-merger price effect in percent (2020Q1–2022Q3). **Bold/green row = selected specification (K=5).**

Appendix B: Model Fit — RMSE and MAE

Table 7.2: Model Fit: RMSE / MAE vs. Observed Prices (Post-Merger Quarters 2020Q1–2022Q3)

	RMSE / MAE (USD)		
Brand / Scope	M1: Logit	M2: Brand-Nested	M3: CLIP-Nested
All	**1.558 / 0.743**	**1.556 / 0.746**	**1.564 / 0.810**
hello	1.579 / 0.836	1.580 / 0.837	1.533 / 0.799
Colgate	1.144 / 0.559	1.142 / 0.560	1.149 / 0.652

Notes: Each cell shows RMSE / MAE in USD against observed transaction prices. Nash (merger) prices are used as the model prediction. "All" = all 847 ASIN × quarter observations in the post-merger period (2020Q1–2022Q3); hello = 35 obs.; Colgate = 172 obs. Pre-merger marginal costs are held fixed in both Nash simulations.

Appendix C: Robustness Checks

Table 7.3: Robustness: Market Size, Calibration Level, and Cost Window

Scenario	λ	α	ρ^*	RSS drop	Neg. MC %	hello Δp	Colgate Δp
M1 baseline	**13.93**	**-0.310**	**0.00**	**0.0%**	**8.6%**	**+1.233%**	**+0.104%**
M1, market size x2	27.86	-0.310	0.00	0.0%	8.4%	+0.634%	+0.054%
M1, market size x5	69.65	-0.309	0.00	0.0%	8.4%	+0.258%	+0.022%
M1, market size x10	139.30	-0.309	0.00	0.0%	8.4%	+0.130%	+0.011%
M3 baseline	**13.93**	**-0.310**	**0.67**	**8.1%**	**1.7%**	**+0.404%**	**+0.018%**
M3, market size x2	27.86	-0.310	0.67	8.1%	1.6%	+0.211%	+0.010%
M3, market size x5	69.65	-0.309	0.67	8.1%	1.6%	+0.087%	+0.004%
M3, market size x10	139.30	-0.309	0.67	8.1%	1.6%	+0.044%	+0.002%
M2, alpha recalibrated at rho*	13.93	-0.212	0.35	3.1%	20.7%	+1.810%	+0.157%
M3, alpha recalibrated at rho*	13.93	-0.111	0.67	8.1%	36.3%	+0.944%	+0.057%
M3, MC from 2019Q4 only	13.93	-0.310	0.67	8.1%	1.7%	+0.440%	+0.022%

Notes: Bold rows are the baseline specifications. **Market size** rows multiply λ (and hence market size M_t) by 2, 5, and 10; α is recalibrated to the -2.62 target at $\rho = 0$ within each scenario. Fit and the negative-cost rate are essentially unchanged; absolute merger effects shrink mechanically as more diversion flows to the outside good, while the M1-to-M3 ratio of hello effects is stable. **Alpha recalibrated at** ρ^* rows hold the baseline profiled ρ^* fixed and rescale α so the model's own implied median elasticity equals -2.62; the resulting small $|\alpha|$ inflates Bertrand markups and drives negative-cost rates far above the plain-logit level, indicating that ASIN-level demand must be more elastic than the aggregate benchmark to rationalize observed prices. **MC from 2019Q4 only** fixes marginal costs at the last pre-merger quarter instead of the pre-merger average.

Appendix D: Equivalence of plain-logit and brand-nested markups

Proposition. Consider nested-logit demand with nesting parameter $\rho \in [0, 1)$ and Bertrand competition, and suppose the nest partition refines the ownership partition: every nest g is wholly owned by a single firm. Then the multiproduct Bertrand markups solving the first-order conditions at given shares are identical to the plain-logit ($\rho = 0$) markups: for every product j of firm f, $p_j - mc_j = 1/(|\alpha|(1 - S_f))$, where $S_f = \sum_{k \in f} s_k$ is the firm's inside share.

Proof. Write $x_k = p_k - mc_k$. Firm f's first-order condition for product j is $s_j + \sum_{k \in f} x_k\, \partial s_k / \partial p_j = 0$. Under plain logit, substituting the share derivatives and dividing by αs_j gives $x_j - \sum_{k \in f} s_k x_k = -1/\alpha$, whose solution is uniform within the firm: $x_j = \mu_f$ with $\mu_f(1 - S_f) = -1/\alpha$. Under nested logit with j in nest $g \subseteq f$, the same substitution gives

$$\frac{x_j}{1-\rho} - \frac{\rho}{1-\rho} \sum_{k \in g} s_{k|g}\, x_k - \sum_{k \in f} s_k x_k = -\frac{1}{\alpha}.$$

Try the uniform candidate $x_k = \mu_f$ for all $k \in f$. Because the entire nest g belongs to firm f, the within-nest shares sum to one, $\sum_{k \in g} s_{k|g} = 1$, so the first two terms collapse to $\mu_f \left(\frac{1}{1-\rho} - \frac{\rho}{1-\rho} \right) = \mu_f$, and the condition reduces to $\mu_f(1 - S_f) = -1/\alpha$ — the plain-logit equation. Since the system $(\Omega \circ \Delta)\,\mathbf{x} = -\mathbf{s}$ is invertible in our data, this uniform solution is the unique solution, and it does not depend on ρ. □

The empirical counterpart is exact: across all 1,164 observations, M1 and M2 marginal costs agree to machine precision. The equivalence breaks whenever a nest spans products of different owners — the CLIP clusters do, which is why M3 recovers different (and far more plausible) costs.